Manipulation-Resistant Oracles via Continuous Peer Prediction

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Abstract. Blockchain oracle networks, including decentralised physical infrastructure networks (DePINs), report physical measurements from a source that can manipulate the underlying process. A carbon project operator, for example, can dry a site before measurement and be paid for soil organic carbon (SOC) that was never stored. Peer prediction covers discrete signals under a strategic source, or continuous reports under an honest source, but not both at once. We close that gap for Gaussian reports. Strict propriety of the continuous ranked probability score (CRPS) settles the reporting stage alone. Our result concerns the source stage. A report-only settlement rule whose honest-source payoff is non-decreasing in the true value identifies and deters source manipulation if and only if the sensors respond to that manipulation by different amounts. When they respond identically, manipulation is observationally equivalent to a genuine rise in SOC, and CRPS translation equivariance leaves every peer score unchanged, so no such rule deters the source. Necessity follows from invariance rather than from the signal enumeration available in the discrete case. The source-stage estimator is ordinary weighted least squares, and the contribution is the characterisation around it. With eight calibrated SOC models on 37,280 real European soil spectra, shift-naive aggregators over-credit by up to 57% under a calibrated moisture perturbation while the joint estimate stays near-unbiased. The guarantee holds on the equilibrium path for non-colluding reporters and inherits the models' own calibration error. The extended version with the full proofs is at `https://rqcker.github.io/crps-wls`.

Keywords: Peer Prediction · Mechanism Design · Source Manipulation · Blockchain Oracles · Carbon Markets · Stochastic Modelling.

1 Introduction

Carbon credit markets pay landowners for storing carbon in soil, and the payment depends on a **S**oil **O**rganic **C**arbon (**SOC**) measurement that the buyer

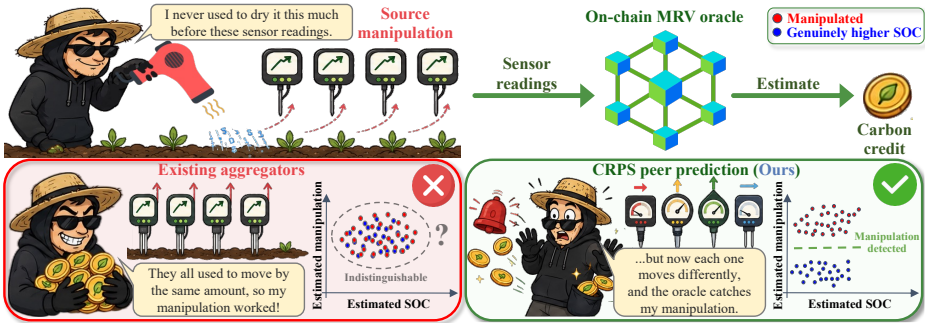


Fig. 1. Sensor diversity separates manipulation from real carbon. Drying the site shifts every sensor's reading. A uniform shift is indistinguishable from genuinely higher carbon (**left**), but when sensor types respond by different amounts the pattern of readings reveals the manipulation (**right**).

cannot independently check [19]. Soil moisture changes the water absorption bands of a spectrum and biases the predicted SOC [17]. Seven of the eight models calibrated in Section 5 read higher on drier spectra, so an operator who dries the site before measurement is paid for carbon that is not there. The party that controls the measurement site is the source of the signal, with both the means and the motive to manipulate it (Figure 1).

This failure is an instance of a broader one. Oracle manipulation has drained hundreds of millions from decentralised finance, including over $110 million from Mango Markets in 2022 [28]. Those attacks corrupt a reported value. Source manipulation works one step earlier: it changes the physical quantity before measurement, so the sensor models report honestly and the oracle relays a true reading of an engineered reality.

Blockchain settles the carbon payment trustlessly but cannot see the soil. A **M**easurement, **R**eporting, and **V**erification (**MRV**) oracle brings the off-chain reading on-chain, and **De**centralised **P**hysical **I**nfrastructure **N**etworks (**DePINs**) such as Open Forest Protocol (OFP) [20], WeatherXM [32], and Boomitra [3] already run them. Each aggregates readings by majority vote, self-consistency, or a centralised model, and none scores the source for honesty. A source that shifts every sensor by the same amount defeats them all, since a uniform shift is indistinguishable from genuinely higher carbon.

Four roles run through the paper. The *source* owns the site and chooses the physical intervention. A *sensor model* turns the resulting measurement into a calibrated Gaussian report about the SOC value. A *reporter* submits that report and collects a peer payment. The *settlement rule* reads the submitted reports, issues credits, and slashes the source collateral. The source and the reporters are the strategic parties; a sensor model is the calibrated pipeline a reporter runs.

Peer prediction scores an agent by how well its report predicts a peer's, so truthful reporting is an equilibrium with no ground truth [16]. Two lines of work approach our setting but do not meet in it. Milionis et al. [15] add a

manipulating source, but only for discrete signals. Richardson and Faltings [22] handle the continuous distributional reports a calibrated model produces, but assume an honest source. SOC measurement needs both at once. Milionis et al. note that their discrete result can be generalised to continuous signals, and we carry out that extension for the Gaussian case.

Strict propriety of the **C**ontinuous **R**anked **P**robability **S**core (**CRPS**) [10] settles the reporting stage. A reporter scored on how well its distribution predicts a peer's report can do no better than state that distribution truthfully. Propriety says nothing about whether the source action is visible in the report law. That question is the source stage. Manipulation enters there as a nuisance direction, and the joint estimate of the SOC value and the manipulation is ordinary weighted least squares. The contribution is the characterisation around that estimate. Identification and deterrence by report-only settlement rules hold exactly when the sensitivity vector is transverse to the all-ones direction. The necessity argument cannot reuse the discrete proof, which enumerates a finite signal alphabet. It rests instead on an invariance. CRPS is translation equivariant, so a common shift of every report leaves all peer scores unchanged. Uniform sensitivities then make manipulation observationally equivalent to a genuine rise in SOC. We call the transversality requirement the **SOC Sensor Diversity Condition** and show that it is the exact dividing line. Diversity is a requirement rather than a modelling convenience, and it is testable, since different model families respond differently to the same physical change. The condition is stated here for one manipulation direction. An adversary free to choose the intervention needs diversity against every feasible direction, the multi-mode condition of Section 4.

Our contributions are as follows:

- We give a peer prediction mechanism for continuous Gaussian reports under an adversarial source, the case left open between discrete strategic-source and continuous honest-source peer prediction.
- We give an if-and-only-if characterisation (Theorem 1): source manipulation is identifiable, and deterrable by report-only rules whose honest-source payoff is non-decreasing in the true value, exactly when the sensors are diverse. When diversity fails, the translation equivariance of CRPS makes manipulation indistinguishable from genuinely higher carbon, and no such rule deters it.
- On 37,280 real European soil spectra with eight calibrated SOC models, shift-naive aggregators over-credit by up to 57% under the calibrated moisture perturbation, while the joint estimate stays near-unbiased and about seven times less sensitive. We report the price too: a higher honest-case error floor, modest detection power, and no paired wet-and-dry validation.

2 Related Work

Peer prediction. Peer prediction mechanisms elicit truthful reports from agents who observe correlated private signals without access to a ground truth. Miller

et al. [16] introduced the peer prediction method, which scores an agent by how well its report predicts a peer's signal under a common prior. It makes truthful reporting an equilibrium, though not the only one, and later work strengthens the guarantee. Schoenebeck and Yu [24] propose the Source and Target **D**ifferential **P**eer **P**rediction (**DPP**) mechanisms, strongly truthful for three agents on a single question, with prior-free guarantees but finite signal reports and an honest source. Zhao et al. [35] apply peer prediction to the blockchain verifier's dilemma for binary signals, and Wang et al. [31] use it for forecast aggregation. Neither models a manipulating source. Zhang et al. [34] propose stochastic dominance truthfulness, a stronger criterion, but establish it for binary signals only. Multi-task peer prediction attains informed truthfulness that resists uninformed and colluding strategies [7,25], but a single physical site offers only one task. None of these works address continuous distributional reports or source manipulation of a physical quantity.

Continuous and adversarial peer prediction. Two lines come closest to our setting but stop short of it. Richardson and Faltings [22] introduce peer neighbourhood mechanisms, which extend peer prediction to arbitrary signal distributions. They achieve strict Bayes–Nash equilibria for continuous and distributional reports, including the Gaussian reports that calibrated SOC models produce, but they assume an honest source. Frongillo et al. [9] study agents who hold more private signals than they report, yet restrict reported outputs to binary values. Milionis et al. [15] add a manipulating source and extend the Source-DPP framework to DePIN settings. Their source identifiability condition is necessary and sufficient for a strictly truthful mechanism, but the signal model is discrete and the proof enumerates a finite alphabet. They note that the result can be generalised to continuous signals, and we carry out that extension for Gaussian signals, replacing the enumeration by the translation equivariance of CRPS [10]. That score sits within the broader theory of property elicitation [14], and Neyman and Roughgarden [18] show it yields minimax-optimal forecast aggregation under truthful reporting. Strict propriety governs the reporting stage in all of this work. What is open is whether the source action is visible in the report law at all.

Robust and Byzantine-resilient aggregation. A separate line defends aggregation against corrupted inputs without modelling incentives. Robust statistics replace the mean with high-breakdown estimators [11], and Byzantine-resilient rules such as Krum [2] and Bulyan [8] filter outlying updates. These assume a corrupted minority and a clean majority, so they suppress a few deviating sensors. Source manipulation breaks that premise: a common shift moves every sensor at once and no clean majority remains.

Soil spectroscopy and moisture compensation. Soil visible-to-near-infrared spectroscopy is an established low-cost proxy for SOC whose accuracy depends on calibration design and acquisition conditions [27]. Moisture is a documented nuisance there, and a line of work removes it before prediction: external parameter orthogonalisation [17,33] and direct standardisation [23] project the water response out of the spectrum, and a trusted moisture reading can enter the

Table 1. Related work compared on the five properties used here. Signal type, report type, source-manipulation modelling, scoring rule and deployment context. CRPS denotes the continuous ranked probability score.

Method	Continuous	Distributional	Source	CRPS	Blockchain
Miller et al. [16]	–	–	–	–	–
Schoenebeck & Yu [24]	–	–	–	–	–
Richardson & Faltings [22]	✓	✓	–	–	–
Milionis et al. [15]	–	–	✓	–	✓
Neyman & Roughgarden [18]	✓	✓	–	✓	–
Frongillo et al. [9]	✓	–	–	–	–
Deployed DePIN oracles [20,32,3]	–	–	–	–	✓
CRPS-WLS Oracle (Ours)	✓	✓	✓	✓	✓

calibration as a covariate. These methods act upstream of the oracle and need reference data, paired wet and dry spectra, or a moisture sensor the source does not control. They reduce a known measurement nuisance. They do not score a party free to choose the measurement condition. The two layers are complementary. Better moisture compensation feeds better calibrated reports into the mechanism studied here.

Blockchain oracles for carbon MRV. As attention to the circular economy and the low-carbon transition grows [26], on-chain carbon verification needs an oracle that is both accurate and manipulation-resistant. Open Forest Protocol [20] votes on a reported value through multi-validator output agreement, WeatherXM [32] scores sensors by self-consistency (its peer-comparison check is planned but not deployed), and Boomitra [3] issues credits from a centralised model without incentive guarantees. Financial oracle designs harden price feeds with gas-efficient or volume-weighted medians [1,4], but neither elicits honest reports from a strategic source, and industry guidance treats oracle security as an economic cost-of-attack condition rather than an absolute guarantee [5]. Cong et al. [6] frame an oracle trilemma among scalability, decentralisation, and truthfulness, and the present mechanism targets its truthfulness corner under the SOC Sensor Diversity Condition. Table 1 positions it against the closest prior results. The settlement layer here is a closed-form rule, not a systems contribution. It does not establish that a sample came from the claimed parcel, enforce the measurement protocol, or govern how the sensitivities and noise levels are calibrated. We report no contract implementation or gas cost.

3 System Model

3.1 Sensor Modelling

Each of $N \geq 2$ sensor models estimates the SOC of a common site, a latent scalar $\theta \in \mathbb{R}$ (in g/kg). We use the flat-prior Gaussian location model as the baseline because it gives the cleanest reporting rule, $m_i = Y_i$. Section 4.3 gives the proper-prior variant, under which identification and deterrence remain exact.

A self-interested source, the party that owns the site and earns carbon credits, chooses a manipulation action $s \in \mathcal{A}_S = \mathbb{R}_{\geq 0}$ before measurement, a physical intervention such as drying the soil, with $s = 0$ the honest source. The scalar action s represents a fixed, calibrated manipulation direction. If the source can choose among several physical interventions, the model gains one sensitivity column per intervention and the relevant condition becomes the multi-mode rank condition stated in Section 4. The one-dimensional model below should therefore be read as local to the calibrated wetting and drying direction used in the experiments. Because s acts on the site, it reaches every sensor model through the same physical event, which is what makes it a single column of the design below. Tampering with individual reports is a different channel: a free per-report shift s_i adds one unknown per report and is not identifiable from one round of N reports without further structure, so it lies outside the guarantee (Section 5.4).

Each sensor model is a calibrated machine learning predictor, and a reporter submits its output. Sensor model i observes:

$$Y_i \;=\; \theta + \alpha_i\, s + \varepsilon_i, \qquad \varepsilon_i \sim \mathcal{N}(0, \sigma_i^2), \tag{1}$$

where $\sigma_i > 0$ is the model's calibrated noise level. For the Gaussian likelihood we take $\boldsymbol{\varepsilon} \sim \mathcal{N}(0, \Sigma)$ with $\Sigma_{ii} = \sigma_i^2$. The errors need not be independent, since the models read a common spectrum, but the displayed peer score uses the diagonal case for clarity. A general covariance is algebraically identical for identification, which depends only on the mean map $(\theta, s) \mapsto \theta\mathbf{1} + s\boldsymbol{\alpha}$, and affects only precision, detection power, and the stake. The coefficient $\alpha_i > 0$ is the **manipulation sensitivity** of sensor model i, and different architectures aggregate spectral information differently, so the α_i need not coincide. We collect the parameters into vectors $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_N)^\top$ and $\boldsymbol{\sigma} = (\sigma_1, \ldots, \sigma_N)^\top$, both common knowledge. The values of $\boldsymbol{\alpha}$ and $\boldsymbol{\sigma}$ are calibration inputs to the settlement rule, not strategic reports. They are public by design, and the estimator and the penalty are built for those public values. Misspecification of $\boldsymbol{\alpha}$ is treated separately in Section 4.

Each reporter states a Gaussian belief about θ, the calibrated output of the models we consider [22]: a pair $R_i = (m_i, v_i)$ encoding $\mathcal{N}(m_i, v_i)$. In the diffuse-prior limit reporter i's posterior for θ given Y_i is $\mathcal{N}(Y_i, \sigma_i^2)$, so truthful reporting is $R_i = (Y_i, \sigma_i^2)$, and then:

$$\mathbb{E}[m_i \mid \theta, s] \;=\; \theta + \alpha_i\, s. \tag{2}$$

Under a proper common Gaussian prior the truthful report is the posterior belief instead, and Section 4.3 gives the fixed de-shrinking map that recovers an unbiased signal proxy before aggregation. We write $R = (R_1, \ldots, R_N)$ for the report profile and $L_{\theta,s}$ for the joint law of R at true value θ and source action s under truthful reporting.

3.2 CRPS Peer Scoring

The mechanism scores reports with the CRPS, the strictly proper scoring rule for full predictive distributions [10]. For a predictive distribution with a cumulative

distribution function F and an observation y,

$$\mathrm{CRPS}(F, y) \;=\; \int_{-\infty}^{\infty} \big(F(z) - \mathbf{1}\{z \geq y\}\big)^2 \, dz. \tag{3}$$

For a Gaussian forecast $\mathcal{N}(m, \sigma^2)$ with $z = (y - m)/\sigma$ this has the closed form:

$$\mathrm{CRPS}\big(\mathcal{N}(m, \sigma^2), y\big) = \sigma\Big[\, z\big(2\Phi(z) - 1\big) + 2\phi(z) - \tfrac{1}{\sqrt{\pi}} \,\Big], \tag{4}$$

where ϕ and Φ are the standard normal density and distribution functions. Reporter i is scored against a peer $j \neq i$ by how well it predicts the peer's observation, in the on-path, flat-prior, diagonal-covariance form used in the baseline theorem. At the candidate equilibrium the source plays $s = 0$, so a reporter whose belief about θ is $\mathcal{N}(m_i, v_i)$ predicts Y_j as $\mathcal{N}(m_i, v_i + \sigma_j^2)$, since Y_j then adds independent noise of the known variance σ_j^2 to θ in (1). The mechanism therefore inflates the reported variance by that known σ_j^2 and scores the result against the peer's reported mean m_j:

$$S_{ij}(R) \;=\; \mathrm{CRPS}\big(\mathcal{N}(m_i,\, v_i + \sigma_j^2),\; m_j\big) \;=\; \kappa\Big[\, d\big(2\Phi(d) - 1\big) + 2\phi(d) - \tfrac{1}{\sqrt{\pi}} \,\Big], \tag{5}$$

where the right-hand closed form specialises (4) to the truthful variance $v_i = \sigma_i^2$, with $\kappa = \sqrt{\sigma_i^2 + \sigma_j^2}$ and $d = (m_i - m_j)/\kappa$. The payment to reporter i is $-\frac{1}{N-1}\sum_{j \neq i} S_{ij}(R)$, so a lower peer CRPS yields a higher payment. The solution concept is the on-path Bayes–Nash equilibrium made precise in Section 4.

3.3 Sensor Diversity

The source is the third player, in the spirit of source differential peer prediction [24,15]. After scoring the reporters, the mechanism aggregates the reports into an SOC estimate that sets the credits and penalises the source by an amount that depends on the reports. For the source to face a strict incentive to stay honest, the mechanism must tell manipulation apart from a genuinely higher SOC value, which is possible only if distinct pairs (θ, s) induce distinct report laws.

Definition 1 (SOC Sensor Diversity). *The ensemble satisfies the* ***SOC Sensor Diversity Condition*** *if the report law identifies the latent value and the source action jointly, that is,* $L_{\theta,s} = L_{\theta',s'}$ *implies* $(\theta, s) = (\theta', s')$.

For the linear Gaussian model of (1) this abstract condition reduces to a transparent algebraic one. Stacking (2) gives the mean vector $\mathbb{E}[\boldsymbol{m}] = A\,(\theta, s)^\top$ with design matrix:

$$A \;=\; [\,\mathbf{1} \;\mid\; \boldsymbol{\alpha}\,] \in \mathbb{R}^{N \times 2}, \qquad \mathbf{1} = (1, \ldots, 1)^\top. \tag{6}$$

Lemma 1 (Algebraic form of diversity). *For the model* (1)*, SOC Sensor Diversity holds if and only if* $\mathrm{rank}(A) = 2$*, the sensitivities* α_i *are not all equal.*

Proof. The variances are common knowledge and do not depend on (θ, s), so $L_{\theta,s}$ is determined by the mean vector $A(\theta, s)^\top$. Diversity is therefore injectivity of $(\theta, s) \mapsto A(\theta, s)^\top$, which holds if and only if the columns $\mathbf{1}$ and $\boldsymbol{\alpha}$ are linearly independent, that is $\operatorname{rank}(A) = 2$, equivalently the α_i are not all equal. □

Diversity therefore states that the manipulation direction $\boldsymbol{\alpha}$ is transverse to the all-ones direction $\mathbf{1}$, along which a change is indistinguishable from a change in θ. This geometric reading drives both directions of the main result, and Proposition 2 formalises it as location invariance. When the α_i are close but not equal, A is ill-conditioned and the variance of $\hat{s}$ grows towards degeneracy, which a deployer counters by adding a sensor model with a distinct sensitivity profile. For a richer feasible-action set the same argument applies to the expanded design $[\,\mathbf{1} \mid \boldsymbol{\alpha}^{(1)} \mid \cdots \mid \boldsymbol{\alpha}^{(k)}\,]$.

4 Equilibrium and Source Deterrence

4.1 Mechanism Construction

The mechanism has two stages: the first scores reporters, the second aggregates their reports and settles the source. The contribution is the incentive characterisation, not the estimator, which is ordinary **W**eighted **L**east **S**quares (**WLS**). CRPS peer scoring elicits truthful reports at the honest-source equilibrium, while the diversity condition of Section 3 decides whether source manipulation is identifiable and deterrable by report-only rules. Theorem 1 separates the two claims.

Reporter stage. Each reporter receives the peer payment defined after (5). Because CRPS is strictly proper and the mechanism scores each reporter's predictive for the peer observation, a reporter minimises its peer CRPS only by reporting that predictive truthfully (Proposition 1).

Source stage. Given the reports, the mechanism solves the WLS problem induced by (2) and (6). With weights $W = \operatorname{diag}(1/\sigma_1^2, \ldots, 1/\sigma_N^2)$ and report means $\boldsymbol{m} = (m_1, \ldots, m_N)^\top$, the estimator is:

$$\big(\hat{\theta},\, \hat{s}\big)^\top \;=\; (A^\top W A)^{-1} A^\top W \boldsymbol{m}, \tag{7}$$

which is defined exactly when $\operatorname{rank}(A) = 2$. The mechanism issues credits proportional to $\hat{\theta}$ and charges the source a penalty $P(\hat{s})$, where P is non-decreasing and non-constant with $P(z) = 0$ for $z \le 0$. The source action is one-sided, $s \ge 0$, since only upward inflation raises the credited aggregate. The source's payoff from action s, net of credits and penalty:

$$U_S(s) \;=\; p\,\mathbb{E}[\hat{\theta} \mid s] \;-\; \mathbb{E}\big[P(\hat{s}) \mid s\big], \tag{8}$$

where $p > 0$ is the credit price. Crediting on $\hat{\theta}$ is deliberate. Under diversity $\hat{\theta}$ is unbiased (Theorem 1), so the credit term does not vary with s and manipulation yields no credit gain. The penalty then makes honest sourcing strictly best rather than merely offsetting a gain. Section 5.4 sizes the stake for the alternative

deployment that credits the more accurate but manipulable precision-weighted mean. We take manipulation to be costless to the source, the adversarial worst case in which the mechanism must deter cheating through scoring alone. An intervention that costs the source energy or labour only makes deterrence easier, so the analysis needs no estimate of that cost.

We adopt the continuous peer prediction setting [22], where the variance inflation in (5) makes truthful reporting proper without a separate posterior-consistency hypothesis. At the candidate equilibrium $s = 0$, so the sensitivities enter only through $\alpha_j s$ and diversity plays no role at the reporter stage. Its role is at the source stage, where it makes s identifiable.

4.2 Main Theorem

The necessary direction rests on a single structural property of CRPS.

Lemma 2 (CRPS translation equivariance). *For every $c \in \mathbb{R}$, every distribution F, and every y,*

$$\mathrm{CRPS}(F^c, y + c) \;=\; \mathrm{CRPS}(F, y), \tag{9}$$

where $F^c(z) = F(z-c)$ is F shifted by c. Consequently, the peer score (5) depends on the report means only through the difference $m_i - m_j$, and a common shift $m_k \mapsto m_k + c$ applied to all reporters leaves every peer score unchanged.

Proof. Substituting $u = z - c$ in (3) gives the identity, and (5) depends on the means only through $d = (m_i - m_j)/\kappa$. □

Proposition 1 (Strict reporter-stage equilibrium). *Under the standing assumptions, with the source playing $s = 0$ and all peers reporting truthfully, reporting $R_i = (Y_i, \sigma_i^2)$ is the unique best response for every reporter. Truthful reporting is therefore a strict Bayes–Nash equilibrium of the reporter stage.*

Proof. Strict propriety [10] uniquely minimises each peer term at reporter i's predictive $\mathcal{N}(Y_i, \sigma_i^2 + \sigma_j^2)$, forcing $m_i = Y_i$ and $v_i = \sigma_i^2$. The peer-specific σ_j^2 cancels, so one report minimises every term at once. □

Scope of the guarantee. The result below is a strict Bayes–Nash statement on the equilibrium path. Three limits follow. First, a source deviation is evaluated holding the reporters' equilibrium strategy fixed, under which the estimator identifies the deviation through $\hat{s}$. We claim no sequential or perfect-Bayesian refinement. If reporters placed positive probability on $s > 0$ and re-optimised, their peer-predictive means would generally differ from the raw Y_i under heterogeneous sensitivities. That could attenuate the pattern that identifies s. Second, truthful reporting is strict against unilateral deviations, not unique. A coalition that controls every reporter and adds a common shift $b\mathbf{1}$ to the reported means raises the credit by b while leaving $\hat{s}$ and, by Lemma 2, every peer score unchanged. That is the collusion form of the degeneracy the theorem rules out on

the source side, and it is why a live deployment needs sealed reports and Sybil-resistant identity. Every partial coalition moves its reports away from its peers and pays a strictly positive peer-CRPS penalty. Third, the impossibility direction restricts the class of settlement rules, as a rule free to punish high reports as such would deter manipulation without paying for carbon.

Definition 2 (Admissible settlement rule). *A settlement rule is* report-only *if the credit it pays and the penalty it charges are functions of the report profile R alone, and* value-monotone *if its honest-source net payoff $V(\theta)$, evaluated under truthful reporting at true value θ with $s = 0$, is non-decreasing in θ. A rule with both properties is* ***admissible****.*

Lemma 3 (Stochastic monotonicity). *If $\hat{s} \mid s \sim \mathcal{N}(s, v_s)$, then $\hat{s}$ is stochastically increasing in s. Hence for any non-decreasing P, $\mathbb{E}[P(\hat{s}) \mid s]$ is non-decreasing in s, and strictly increasing whenever P is non-constant.*

Theorem 1 (Source deterrence if and only if sensor diversity). *In the flat-prior Gaussian reporting model, CRPS peer scoring makes truthful reporting a strict best response at the honest-source equilibrium. Fix truthful reports by non-colluding reporters. There is then a penalty P for which $s = 0$ is a strict source best response if and only if SOC Sensor Diversity (Definition 1) holds. Necessity holds for every admissible settlement rule (Definition 2) and uses no property of the scoring rule. Under a proper common Gaussian prior the same conclusion follows after the de-shrinking transformation of Section 4.3.*

Proof (sketch). Sufficiency. Under diversity $\operatorname{rank}(A) = 2$ (Lemma 1), so the WLS estimator (7) is unbiased and $\hat{s}$ identifies any source deviation taken against the fixed equilibrium reporting strategy. The credit term in (8) is then constant in s, and a penalty non-decreasing in $\hat{s}$ makes $s = 0$ the strict best response by Lemma 3. *Necessity.* If the sensitivities are all equal to $\bar{\alpha}$, source action s at true value θ_0 and an honest source at the higher true value $\theta_1 = \theta_0 + \bar{\alpha}s$ induce the identical report law, $L_{\theta_0,s} = L_{\theta_1,0}$. A report-only rule pays the same credit and the same penalty in both scenarios, and by translation equivariance (Lemma 2) every peer score is unchanged as well. Value monotonicity then gives $U_S(s)|_{\theta_0} = V(\theta_0 + \bar{\alpha}s) \geq V(\theta_0) = U_S(0)|_{\theta_0}$ for every $s > 0$, so no penalty makes $s = 0$ strict. Neither step invokes a property of the score, so the impossibility covers every scoring rule. The full argument is on the project webpage. □

The necessity argument has no finite analogue: the indistinguishability $L_{\theta_0,s} = L_{\theta_1,0}$ holds in the continuous model directly, so the characterisation follows by a different route than the discrete enumeration of [15] rather than as a corollary of it. This is what strict propriety does not supply. Propriety fixes what a reporter should say, whereas the question here is whether the source action is present in the report law at all.

Extensions. The characterisation generalises to k manipulation modes, where strict truthfulness needs $\operatorname{rank}(A) = k + 1$, and to a known noise covariance Σ, under which the diagonal estimator stays exactly unbiased and only $\operatorname{Var}(\hat{s}) =$

$c_s^\top \Sigma\, c_s$ changes. A sensitivity drift biases the credit only through its non-proportional part. Sufficiency uses only strict propriety and mean elicitation, so the logarithmic and energy scores qualify, while necessity uses no property of the score at all. The project webpage carries the details.

4.3 Invariance and Aggregation

Detecting manipulation is an invariant inference problem, which is why the prior on θ plays no role in it. The location group acts on report means by $g_c\, \boldsymbol{m} = \boldsymbol{m} + c\,\mathbf{1}$, and shifting $\theta \mapsto \theta + c$ shifts every truthful mean by g_c through (2), so θ is the group parameter and the manipulation s is the invariant part.

Proposition 2 (Prior-free manipulation detection). *Let $\Pi = \mathbf{I} - \frac{1}{N}\mathbf{1}\mathbf{1}^\top$ and let $T = \Pi \boldsymbol{m}$ be the contrast vector of the report means, the maximal location invariant. Under truthful reporting $T \sim \mathcal{N}(s\,\Pi\boldsymbol{\alpha},\, \Pi\Sigma\Pi)$, a law free of θ, and the row $c_s^\top$ of (7) satisfies $c_s^\top \mathbf{1} = 0$ and $c_s^\top \boldsymbol{\alpha} = 1$, so $\hat{s} = c_s^\top T$. The law of $\hat{s}$, and with it detection and deterrence, is therefore invariant to θ and to any prior placed on it. Manipulation is identifiable from T if and only if $\Pi\boldsymbol{\alpha} \neq 0$, the condition of Lemma 1.*

Only $\hat{\theta}$ is equivariant rather than invariant: it tracks $\mathbf{1}$ and shrinks towards the prior mean under a proper prior. That shrinkage never reaches $\hat{s}$, which is what makes deterrence indifferent to the source knowing θ. Under a proper common prior $\theta \sim \mathcal{N}(\mu_0, \tau^2)$ strict propriety elicits $m_i = (1-\beta_i)\mu_0 + \beta_i Y_i$ with $\beta_i = \sigma_i^{-2}/(\tau^{-2} + \sigma_i^{-2})$, and de-shrinking to $\hat{Y}_i = (m_i - (1-\beta_i)\mu_0)/\beta_i$ before (7) restores $\mathbb{E}[\hat{Y}_i] = \theta + \alpha_i s$ with variance σ_i^2. The de-shrink is a fixed affine map, so identification, the diversity condition, and the necessity argument hold for every finite τ^2.

The estimator (7) reuses the calibrated forecasts that the incentive layer scores. Its honest-case limit is a known optimum. Dropping the source column from (7) gives the precision-weighted mean with $w_i = 1/\sigma_i^2$. That is the minimum-variance unbiased linear aggregate by the Gauss–Markov theorem, and the minimax-optimal pooling for the quadratic loss underlying CRPS [18]. The full estimator does not attain that minimum, since identifying s needs the second column of A. That is the price of robustness, and Section 5 measures it.

On-chain settlement. The mechanism is a deterministic, closed-form function of the reports, so it suits direct smart-contract execution. Reporters commit their reports, and the contract evaluates $(\hat{\theta}, \hat{s})$ and the peer payments, credits $\hat{\theta}$, and slashes the source collateral when $\hat{s}$ exceeds the threshold. The published (σ_i, α_i) also let it reject an ensemble whose precision-weighted sensitivity variance falls below a floor, which turns diversity into a procurement requirement. We stop at the rule itself, and claim no contribution on contract implementation, gas or fixed-point error, report authentication, or the governance that recalibrates α and σ.

Table 2. Sensor diversity holds empirically. SOC benchmark on mineral LUCAS vis-NIR (37,280 samples, five-fold cross-validation), with each model's measured noise σ_i and sensitivity α_i, which span more than two orders of magnitude. Best R^2 and lowest σ_i in **bold**, second in underline.

Model	R^2	σ_i (g/kg)	α_i
CNN	**0.505**	**13.22**	1.000
Random forest	**0.505**	13.23	0.418
Vision Transformer	0.491	13.41	0.538
Swin transformer [12]	0.471	13.66	0.705
Stacked autoencoder [29]	0.443	14.02	0.716
SSL-SoilNet [13]	0.396	14.60	0.202
Ridge (linear)	0.387	14.72	0.665
CNN-LSTM-ECA [30]	0.084	17.98	0.004

5 Experiments

5.1 Experimental Setup

The reporters are instantiated by eight machine learning models trained to predict SOC from real visible-to-near-infrared (vis-NIR) reflectance spectra. They stand in for the calibrated pipelines that reporters would run. The experiments therefore test the statistical layer, source identifiability and the invariance of the credited value, and not the strategic behaviour that Theorem 1 governs. We use the LUCAS subset of the Open Soil Spectral Library, which links each spectrum to a laboratory organic carbon (OC) measurement [21], and keep the 37,280 mineral soils with OC at most 100 g/kg, the carbon-credit deployment domain. This excludes extreme-carbon organic soils, a separate regime with much larger prediction variance. Each spectrum is resampled to a 10 nm grid over 400–2450 nm (206 bands). The eight models are a ridge linear baseline, a random forest, a one-dimensional convolutional network, a spectral vision transformer, SSL-SoilNet [13], a CNN-LSTM with efficient channel attention [30], a shifted-window (Swin) transformer [12], and a stacked autoencoder network [29].

Two parameters of the model in Section 3 are measured, not assumed. The noise level σ_i is each model's five-fold cross-validation root mean squared error. The sensitivity α_i is each model's SOC-prediction response to a soil-moisture perturbation of the spectrum, normalised so the largest is one. Table 2 reports both, with the prediction accuracy (R^2). The measured α_i span 0.004 to 1.000, so the SOC Sensor Diversity Condition holds empirically rather than by assumption.

The moisture perturbation models wetting as added water absorption, darkening the spectrum and deepening the O-H bands at 1450 and 1940 nm. It is built from those known absorption bands rather than from measured wet-and-dry pairs of the same soils, which would be a stronger test and which no open dataset currently supplies. Seven of the eight models predict lower OC under wetting, so credit inflation comes from presenting drier soil. The sign does not affect the mechanism, which depends only on the spread of α_i. A source able to pick its intervention is treated in Section 5.4.

The mechanism experiments bootstrap ground truth θ from the LUCAS 2018 mineral OC values (17,231 samples, median 19.9 g/kg), and source manipulation is a fraction of θ. We report the systematic bias $\mathbb{E}[(\hat{\theta} - \theta)/\theta]$, the over-crediting that manipulation induces, and the mean absolute relative error $\mathrm{MARE} = \mathbb{E}[|\hat{\theta} - \theta|/\theta]$, which also includes random error. Each point averages 1,000 draws over five seeds, ten for the significance test.

Robustness to the weakest sensor model. CNN-LSTM-ECA has near-zero sensitivity and anchors the low end of the spread, so we check how much rests on it. Dropping it leaves sensitivities spanning 0.202 to 1.000, so the design stays full rank. The manipulation estimate only becomes noisier, its standard deviation rising from 18.1 to 22.3 g/kg and detection power at full manipulation falling from 29% to 23%.

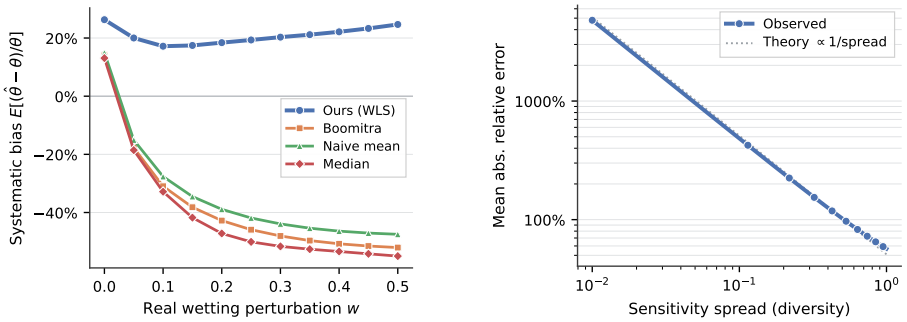
5.2 Manipulation Resistance

At the honest-source equilibrium the peer CRPS of each of the eight reporters, computed analytically from (5), is uniquely minimised at zero deviation from the truthful report, which confirms Proposition 1 on the measured, heterogeneous sensitivities.

Systematic bias. On the parametric model the WLS estimate holds a bias of -1.3% across the whole manipulation range, while the precision-weighted and mean aggregators over-credit by up to $+57\%$ and output agreement reaches $+31\%$ (Table 3). That is the quantity that matters for a carbon market, since systematic over-crediting is fraud whereas random error averages out across parcels. The MARE is flat at 56.8% under diverse sensitivities but grows from 24.4% to 101.8% under uniform ones, and its level is the noise floor of vis-NIR SOC prediction rather than a property of the manipulation.

Trained models on perturbed spectra. We now drop the parametric $\theta + \alpha_i s$ model and run the eight trained models directly on real LUCAS spectra perturbed by the wetting operator (train 29,824, test 7,456). Two facts emerge. First, the models are biased predictors: on clean spectra they over-predict OC by 7% to 27% (one under-predicts by 5%), so any aggregate inherits a calibration bias. Second, the manipulation invariance survives on these real nonlinear responses (Figure 2(a)): the precision-weighted, mean, and median aggregators collapse from about $+14\%$ to between -47% and -55%, while the WLS estimate stays within a 9-point band ($+17\%$ to $+26\%$), about seven times less sensitive. The residual $\approx 20\%$ level is the models' own calibration bias, not the manipulation. The mechanism therefore removes the manipulation bias but inherits the calibration bias. The guarantee is relative (manipulation invariance) rather than absolute (accuracy).

Sensor diversity. Figure 2(b) interpolates the sensitivities from uniform to the measured diverse set under twenty percent manipulation. As the spread approaches zero the error diverges, from 56.8% at the diverse setting to over 4800% near the uniform limit, the empirical boundary of Theorem 1.



(a) (b)

Fig. 2. Invariance holds on real spectra; error diverges as diversity vanishes. **(a)** Systematic bias of the eight trained models on real perturbed spectra; the aggregation rules collapse as wetting grows while WLS stays in a narrow band whose nonzero level is the models' own calibration bias. **(b)** Estimation error versus sensitivity spread at twenty percent manipulation, the empirical form of $v_s \to \infty$.

5.3 Aggregation-Rule Comparison

Table 3 compares the WLS mechanism against seven aggregation rules. Each is the core aggregation rule of a deployed system (precision-weighted mean for Boomitra, outlier removal and peer comparison for WeatherXM, output agreement for OFP), abstracted from its full pipeline. The rows are not evaluations of those pipelines. The comparison isolates the aggregation rule because that is what governs manipulation resistance. Spectral moisture compensation and direct moisture sensing sit upstream of the oracle and are complementary rather than competing (Section 2). At full manipulation six of the seven rules overcredit by more than half and the seventh by 31%, while WLS does not, and the absolute errors converge near the noise floor. A paired one-sided t-test and Wilcoxon signed-rank test over ten seeds find WLS significantly more accurate than six of the seven, with t-test p from 3×10^{-6} to 5×10^{-3}, the Wilcoxon p at its ten-seed floor 9.8×10^{-4} for all but two rows, and every mean absolute relative error inside a $\pm$2-point 95% confidence interval. At zero manipulation the simple aggregators are more accurate, since separating the source costs efficiency when there is nothing to separate. We report that trade-off rather than hide it. Table 3 traces the frontier. MARE_0 is the honest-source error, the variance cost, and bias_1 is the manipulation bias. WLS sits at high MARE_0 and near-zero bias. The aggregators sit at low MARE_0 and large bias.

Estimator versus incentives. The bias gap in Table 3 is a property of the estimator, not of the payments. Any estimator that fits (θ, s) jointly inherits the same unbiasedness once diversity holds: Huber and ℓ_1 regressions on the same design hold a credit bias of -1.5% against the -1.3% of WLS, so the invariance follows from modelling s and not from the loss or the payments. The peer scoring and the penalty do a separate job. They make truthful reporting and

Table 3. Shift-naive aggregation over-credits; the joint estimate does not. Mean absolute relative error at zero (MARE_0) and full (MARE_1) manipulation, systematic bias at full manipulation (bias_1), and the paired one-sided t-test p against our method over ten seeds. Best in **bold**, second in underline.

Method	MARE_0	MARE_1	bias_1	t-test p
Output agreement (OFP)	49%	60%	<u>+31%</u>	4.4×10^{-5}
Precision-weighted (Boomitra)	**24%**	61%	+57%	9.3×10^{-5}
Outlier removal (WeatherXM)	28%	61%	+56%	1.1×10^{-5}
Peer comparison (WeatherXM)	31%	62%	+56%	2.8×10^{-6}
Sample median	28%	61%	+55%	3.2×10^{-5}
CRPS-reweighted mean	<u>25%</u>	<u>58%</u>	+54%	4.5×10^{-3}
Unweighted mean	<u>25%</u>	<u>58%</u>	+53%	0.04
WLS **(Ours)**	57%	**57%**	**−1.3%**	n/a

$s = 0$ a strict equilibrium (Theorem 1), so a rational source gains nothing from manipulating at all, and the detected $\hat{s}$ supports the slashing analysed below.

Robustness. Several departures from the idealised model leave the invariance intact, with the full sweeps on the project webpage. Scaling the ensemble from three to twenty reporters lowers the estimation error and raises detection power at fifty percent manipulation from 10% to 17%. Student-t, lognormal, and uniform noise of the same scale leave the MARE flat between 50% and 57%, as does a fifty percent error in the assumed σ_i, since the theory needs only mean-zero noise. Correlated residuals, which the eight models incur by reading a common spectrum, sharpen detection rather than weakening it, since $\hat{s}$ is a contrast that cancels the common component: detection power at fifty percent manipulation rises from 14% at $\rho = 0$ to 47% at $\rho = 0.9$. The independent-noise figures are therefore the conservative case.

5.4 Security Analysis

The credit rule is a design choice, and the stake exists only for one side of it. Crediting on the debiased estimate $\hat{\theta}$ removes the manipulation gain entirely (Theorem 1), so honest sourcing is strict with no stake, at the cost of accuracy: $\hat{\theta}$ inherits the 57% error floor of Table 3 against 24% for the precision-weighted mean. A deployer who values honest-case accuracy may instead credit on the precision-weighted mean of Section 4.3 and restore resistance with a stake M slashed when $\hat{s} > \delta$.

Detector variance and stake. The manipulation estimate $\hat{s} = c_s^\top \boldsymbol{m}$, with c_s the second row of $(A^\top W A)^{-1} A^\top W$, satisfies $\hat{s} \mid s \sim \mathcal{N}(s, v_s)$ with $v_s = (\sum_i w_i(\alpha_i - \bar{\alpha}_w)^2)^{-1}$, $w_i = \sigma_i^{-2}$ and $\bar{\alpha}_w = \sum_i w_i \alpha_i / \sum_i w_i$, so detection sharpens with the spread of the α_i and $v_s \to \infty$ as that spread vanishes. A one-sided test at false-alarm rate a rejects honest sourcing when $\hat{s} > z_{1-a}\sqrt{v_s}$, giv-

ing power $P_{\text{detect}}(s) = \Phi(s/\sqrt{v_s} - z_{1-a})$. Manipulation is then deterred when $(P_{\text{detect}}(s) - a)M \geq G(s)$ for the precision-weighted credit gain $G(s) = p\,\bar{\alpha}_w s$, hence a minimum stake $M^\star(s) = G(s)/(P_{\text{detect}}(s) - a)$. The deterrent is the detector's power above its size, not its raw power. As the spread vanishes $P_{\text{detect}} \to a$ and $M^\star \to \infty$: no finite stake deters without taxing honest sources equally. That is the economic form of the necessity direction. With $\sqrt{v_s} = 18.1$ g/kg at the median value $\theta = 19.9$ g/kg and $a = 5\%$, detection power is 7.7%, 13.7%, and 29.4% at manipulations of 20%, 50%, and 100% of θ, and $M^\star/G$ falls from 36.8× through 11.5× to 4.1×. At \$15 per tonne of carbon dioxide on one hectare the median parcel earns an honest source \$4,269 per round, a twenty percent manipulation gains $G = \$483$, and the minimum deterring stake is \$17,785. The stake is an economic attack-cost condition and a statistical rather than a cryptographic one: it raises the cost of manipulation without making it infeasible.

Scope and limits. Four limits bound the guarantee, and none is repaired by a larger stake. The detector is statistically thin. The standard deviation of $\hat{s}$ is about 18 g/kg against a median value of 20, so detection power is modest and the stake layer needs the worst-case gain G estimated accurately. The no-bias property holds along $\boldsymbol{\alpha}$, calibrated from a synthetic wetting operator on a single survey. A source free to choose its intervention could pick one whose response is near-uniform, close to span($\mathbf{1}$). That drives the ensemble towards the rank-deficiency boundary that only diversity against every feasible direction rules out. Whether the measured sensitivities are stable across seasons and soil textures needs longitudinal data and paired wet-and-dry field measurements that we do not have. Finally, a coalition of all eight reporters adding a common shift pays no peer penalty and drives $\hat{s}$ to zero, whereas every partial coalition pays a strictly positive peer-CRPS penalty, so a live oracle needs a settlement layer that seals reports and preserves a diversity floor under quorum selection.

6 Conclusion

We extended source-manipulation-resistant peer prediction to continuous Gaussian reports. Strict propriety of CRPS settles only what a reporter should say. The result here is about the source. By translation equivariance, an admissible report-only settlement rule identifies and deters source manipulation if and only if the sensor sensitivities differ, a condition that holds on eight real SOC models. The source-stage estimator is ordinary weighted least squares, and the characterisation around it is what the paper contributes. The guarantee is limited. It holds on the equilibrium path, assumes non-colluding reporters, and is calibrated along one manipulation direction measured from a synthetic wetting operator. It delivers manipulation invariance rather than absolute accuracy. Sensor diversity is therefore a structural requirement for honest sourcing. Paired wet-and-dry validation, diversity against an adversary that chooses its intervention, and a settlement layer that seals reports are left to future work.

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A Deferred Proofs

This appendix gives the full proofs of the results stated in the main text: Theorem 1, Lemma 2, Proposition 1, Lemma 3, the honest-case aggregation statement of Section 4.3, the score-free impossibility (Proposition 3), and the prior-free detection property (Proposition 2).

Proof (of Lemma 2). Substituting $u = z - c$ in (3),

$$\begin{aligned}\mathrm{CRPS}(F^c, y+c) &= \int \big(F(z-c) - \mathbf{1}\{z \geq y+c\}\big)^2 dz \\ &= \int \big(F(u) - \mathbf{1}\{u \geq y\}\big)^2 du \\ &= \mathrm{CRPS}(F, y).\end{aligned}$$

In (5) the score is a function of $d = (m_i - m_j)/\kappa$ and of σ_i, σ_j alone, so it is invariant under $m_i \mapsto m_i + c$, $m_j \mapsto m_j + c$. □

Proof (of Proposition 1). Fix truthful reporting by all peers $j \neq i$ and the equilibrium belief $s = 0$. Sensor i chooses (m_i, v_i) to minimise $\sum_{j \neq i} \mathbb{E}\big[\mathrm{CRPS}\big(\mathcal{N}(m_i,\, v_i + \sigma_j^2),\, Y_j\big) \mid Y_i\big]$. Because CRPS is strictly proper [10], each term is uniquely minimised when the scored distribution equals sensor i's posterior predictive for Y_j, which is $\mathcal{N}(Y_i,\, \sigma_i^2 + \sigma_j^2)$. This forces $m_i = Y_i$ and $v_i + \sigma_j^2 = \sigma_i^2 + \sigma_j^2$, so $v_i = \sigma_i^2$. The peer-specific σ_j^2 cancels, so the single report (Y_i, σ_i^2) minimises every term at once and is the unique minimiser of the sum. Uniqueness makes the equilibrium strict. □

Proof (of Lemma 3). For $\hat{s} \mid s \sim \mathcal{N}(s, v_s)$ the law is a location shift in s, so $\Pr(\hat{s} > x \mid s) = \Phi\big((s - x)/\sqrt{v_s}\big)$ is strictly increasing in s for every x. Hence $\hat{s}$ is strictly stochastically increasing in s, and by first-order stochastic dominance $\mathbb{E}[P(\hat{s}) \mid s]$ is non-decreasing in s for every non-decreasing P, strictly so when P is non-constant. The argument uses no smoothness of P, so it covers the threshold slashing rule. □

Proof (of the honest-case aggregation statement, Section 4.3). With known σ_i and unbiased Gaussian means, inverse-variance weighting is the minimum-variance unbiased linear combination by the Gauss–Markov theorem, and the minimax-optimal aggregate of [18]. Setting the design matrix to the single column $\mathbf{1}$ in (7) yields this weighting. □

Proof (of Theorem 1). Sufficiency ($\Leftarrow$). Assume diversity. By Lemma 1, $\mathrm{rank}(A) = 2$, so the WLS estimator (7) is well defined. By Proposition 1 truthful reporting is a strict best response at the equilibrium action $s = 0$, and a sensor's report equals its observation for any realised s. Under the equilibrium reporting strategy, source deviations change realised observations to $Y_i = \theta + \alpha_i s + \varepsilon_i$, and sensors continue to report those realised observations. Under truthful reporting $\mathbb{E}[\boldsymbol{m}] = A(\theta, s)^\top$, so the WLS estimator is unbiased, $\mathbb{E}[\hat{\theta} \mid s] = \theta$ and $\mathbb{E}[\hat{s} \mid s] = s$,

and $\hat{s}$ identifies any source deviation under that fixed equilibrium reporting strategy. It remains to give the source a strict incentive to play $s = 0$. The credit term in (8) is constant in s because $\mathbb{E}[\hat{\theta} \mid s] = \theta$. By Lemma 3, $\mathbb{E}[P(\hat{s}) \mid s]$ is strictly increasing in s, so on the one-sided action set $s \geq 0$ it is minimised at $s = 0$. Hence $U_S(0) > U_S(s)$ for all $s > 0$, and $s = 0$ is the strict best response. Truthful reporting together with $s = 0$ is a strict Bayes–Nash equilibrium.

Necessity ($\Rightarrow$). We argue the contrapositive. Suppose diversity fails. By Lemma 1, the α_i are all equal to a common value $\bar{\alpha}$. Thus $\boldsymbol{\alpha} = \bar{\alpha}\,\mathbf{1}$, and source action s shifts every report mean by the same amount $\bar{\alpha}s$. Fix any true value θ_0. Under truthful reporting and source action s, the report means have distribution $m_i \sim \mathcal{N}(\theta_0 + \bar{\alpha}s,\, \sigma_i^2)$. This is exactly the profile produced by an honest source ($s' = 0$) at the higher true value $\theta_1 = \theta_0 + \bar{\alpha}s$. The two scenarios induce the identical report law, $L_{\theta_0,s} = L_{\theta_1,0}$. Two consequences follow. First, by Lemma 2 the shift is common to all sensors, so every peer score is unchanged. The sensor payments carry no information about s. Second, any mechanism is a function of the report profile alone. Its credit and penalty terms therefore take the same values in the two scenarios. Write $V(t) := U_S(0)\big|_{\theta=t}$ for the honest-source payoff at true value t. For the payment rules considered here, V is strictly increasing without further assumptions: the credit pays p times a location-equivariant aggregate of the report means (the estimator (7) where defined, with the precision-weighted mean as its honest-case limit), and the penalty reads the reports only through a contrast, whose law is free of θ (Proposition 2), so $V(t) = p\,t - \text{const}$. The source payoff satisfies $U_S(s)\big|_{\theta_0} = V(\theta_0 + \bar{\alpha}s) > V(\theta_0) = U_S(0)\big|_{\theta_0}$ for all $s > 0$. Manipulation is therefore strictly profitable, so $s = 0$ cannot be a strict best response, and no penalty P yields a strict equilibrium. Proposition 3 extends the impossibility to every report-based mechanism whose honest-source net payoff is non-decreasing in the true value. □

Proposition 3 (Score-free impossibility for admissible rules). *Suppose* $\boldsymbol{\alpha} \parallel \mathbf{1}$. *For every scoring rule and every admissible settlement rule in the sense of Definition 2, that is every report-only rule whose honest-source net payoff V is non-decreasing in the true value, manipulation is weakly profitable, so $s = 0$ is not a strict best response. Without value monotonicity the statement fails: a rule that penalises high reported values as such deters manipulation, but it no longer pays for carbon.*

Proof. When $\boldsymbol{\alpha} \parallel \mathbf{1}$ the report laws satisfy $L_{\theta_0,s} = L_{\theta_0+\bar{\alpha}s,0}$, shown in the proof of Theorem 1. A report-only mechanism assigns the two scenarios identical credits and penalties, so the source payoff is $V(\theta_0 + \bar{\alpha}s) \geq V(\theta_0)$ by monotonicity. Sufficiency, by contrast, needs only strict propriety and mean elicitation, so the CRPS, the logarithmic score, and the multivariate energy score all qualify, the last extending the characterisation to vector-valued reports. □

Proof (of Proposition 2). T is the maximal invariant of the location group. Its mean is $\Pi\,\mathbb{E}[\boldsymbol{m}] = \Pi(\theta\mathbf{1} + s\boldsymbol{\alpha}) = s\,\Pi\boldsymbol{\alpha}$ since $\Pi\mathbf{1} = 0$, and its covariance does not involve (θ, s). The normal equations of (7) give $c_s^\top A = (0, 1)$, that is $c_s^\top \mathbf{1} = 0$ and $c_s^\top \boldsymbol{\alpha} = 1$. From $c_s \perp \mathbf{1}$ we get $\Pi c_s = c_s$ and $\hat{s} = c_s^\top \boldsymbol{m} = (\Pi c_s)^\top \boldsymbol{m} = c_s^\top T$.

The mean $s\,\Pi\boldsymbol{\alpha}$ separates the values of s exactly when $\boldsymbol{\alpha} \notin \operatorname{span}(\mathbf{1})$, which is Lemma 1. □

B Stake Sizing

This appendix derives the collateral required for the high-efficiency precision-weighted-credit deployment discussed in Section 5.4, and makes explicit its dependence on the sensitivity spread, the sensor noise, and the detector error rates, the connection summarised in Section 5.

Detector variance. The manipulation estimate is $\hat{s} = c_s^\top \boldsymbol{m}$, where c_s is the second row of $(A^\top W A)^{-1} A^\top W$. Under truthful reporting $\hat{s} \mid s \sim \mathcal{N}(s, v_s)$ with

$$\begin{aligned} v_s \;&=\; \big[(A^\top W A)^{-1}\big]_{22} \;=\; \frac{1}{\sum_i w_i\,(\alpha_i - \bar{\alpha}_w)^2}, \\ w_i = \tfrac{1}{\sigma_i^2}, \qquad \bar{\alpha}_w &= \frac{\sum_i w_i \alpha_i}{\sum_i w_i}. \end{aligned} \tag{10}$$

The denominator is the precision-weighted variance of the sensitivities, so detection sharpens with the spread of the α_i and with lower noise, and $v_s \to \infty$ as the spread vanishes (the uniform-sensitivity limit).

Detection power. A one-sided test rejects honest sourcing when $\hat{s} > \delta$. Fixing the false-alarm (Type I) rate at a sets $\delta = z_{1-a}\sqrt{v_s}$, and the detection power (one minus the Type II rate) at manipulation s is

$$P_{\text{detect}}(s) \;=\; \Pr(\hat{s} > \delta \mid s) \;=\; \Phi\Big(\frac{s}{\sqrt{v_s}} - z_{1-a}\Big). \tag{11}$$

Deterring stakes. Section 5.4 gives the deterrence condition $(P_{\text{detect}}(s) - a)\,M \geq G(s)$ for the precision-weighted credit gain $G(s) = p\bar{\alpha}_w s$. Substituting (11) gives the minimum stake in closed form,

$$M^\star(s) = \frac{G(s)}{P_{\text{detect}}(s) - a} = \frac{p\bar{\alpha}_w s}{\Phi\big(s/\sqrt{v_s} - z_{1-a}\big) - a}. \tag{12}$$

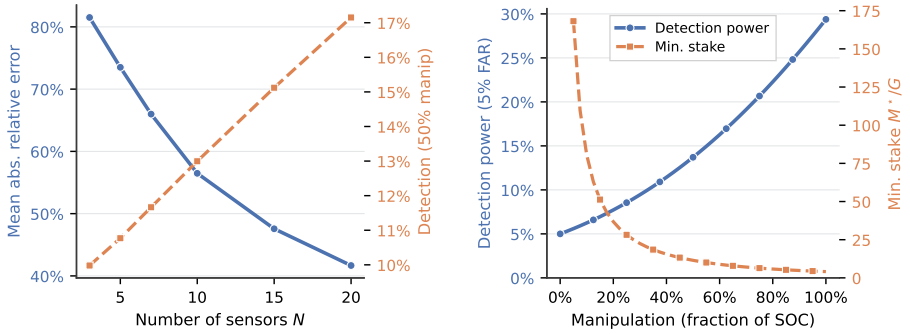
The deterrent is the detector's power above its size, $P_{\text{detect}}(s) - a$, not its raw power. As the sensitivity spread vanishes, $v_s \to \infty$ and $P_{\text{detect}}(s) \to a$, so $M^\star \to \infty$: no finite stake deters without taxing honest sources equally, the economic form of the necessity direction.

The detection power and the minimum stake at the median value $\theta = 19.9$ g/kg, with $\sqrt{v_s} = 18.1$ g/kg and a 5% false-alarm rate, follow from (11)–(12) and are reported in Table 4.

Figure 3 reports the detection power and stake versus manipulation in panel (b), and the gain in accuracy and detection from a larger ensemble in panel (a).

Table 4. The minimum deterring stake falls from 36.8× to 4.1× the gain as manipulation grows. Detection power and minimum stake at the median SOC value ($\theta = 19.9$ g/kg, 5% false-alarm rate), with $\sqrt{v_s} = 18.1$ g/kg.

Manipulation s/θ	20%	50%	100%
Detection power P_{detect}	7.7%	13.7%	29.4%
Minimum stake $M^\star/G$	36.8×	11.5×	4.1×



(a) (b)

Fig. 3. Scaling and stake. (a) Estimation error (left axis) and detection power at fifty percent manipulation (right axis) versus the number of sensors; the honest and manipulated errors coincide. **(b)** Detection power (left axis) and minimum deterring stake $M^\star/G$ (right axis) versus manipulation for diverse sensors; uniform sensors give no power above the false-alarm rate, so no finite stake selectively deters them.

C Reporter-Stage Robustness under Unknown Covariance

Section 4 uses mean reports that are truthful at the flat-prior honest-source equilibrium. The following lemma isolates the limited robustness property needed below: misspecifying the scored variance does not move the CRPS-optimal mean away from the conditional mean of the target distribution.

Lemma 4 (Mean pinning). *Let $Y \sim \mathcal{N}(\mu, \tau^2)$ and fix any $\omega > 0$. The expected score $h(m) = \mathbb{E}_Y[\mathrm{CRPS}(\mathcal{N}(m, \omega^2), Y)]$ is strictly convex in m with the unique minimiser $m = \mu$, independent of ω.*

Proof. The Gaussian CRPS is $\mathrm{CRPS}(\mathcal{N}(m, \omega^2), y) = \omega\, g((y-m)/\omega)$ with $g(z) = z(2\Phi(z) - 1) + 2\phi(z) - \pi^{-1/2}$, so $g'(z) = 2\Phi(z) - 1$. Hence $h'(m) = -\mathbb{E}_Y[g'((Y - m)/\omega)] = 1 - 2\,\mathbb{E}_Y[\Phi((Y-m)/\omega)]$. With an independent $Z \sim \mathcal{N}(0, 1)$, $\mathbb{E}_Y[\Phi((Y - m)/\omega)] = \Pr(m \le Y - \omega Z) = \Phi\big((\mu - m)/\sqrt{\tau^2 + \omega^2}\big)$, so $h'(m) = 2\Phi\big((m - \mu)/\sqrt{\tau^2 + \omega^2}\big) - 1$ and $h''(m) = 2\phi\big((m - \mu)/\sqrt{\tau^2 + \omega^2}\big)/\sqrt{\tau^2 + \omega^2} > 0$. Thus h is strictly convex and $h'(m) = 0$ exactly at $m = \mu$. □

At the reporter stage the target variable in Lemma 4 is the peer observation Y_j, conditional on sensor i's information. The lemma therefore says that a misspecified scored variance does not move the optimal reported mean away from the conditional mean $\mathbb{E}[Y_j \mid Y_i]$. In the flat-prior honest-source model this conditional mean is Y_i. Hence, in that baseline case, variance-inflation errors leave the elicited mean pinned at $m_i = Y_i$. This is the only sense in which the mean report is robust to variance misspecification: Lemma 4 does not assert that Y_i is optimal under arbitrary beliefs about source manipulation or under an arbitrary prior. In those cases the optimal mean is the appropriate posterior-predictive mean.

With known covariance Σ, sensor i's posterior predictive distribution for peer j in the diffuse-prior honest-source model is

$$Y_j \mid Y_i \sim N(Y_i,\, \mathrm{Var}(Y_j - Y_i)) = N\big(Y_i,\, \sigma_i^2 + \sigma_j^2 - 2\Sigma_{ij}\big)\,. \tag{13}$$

Thus the covariance-corrected version of the peer score replaces the independent inflation in (5) by the pre-calibrated variance $\sigma_i^2 + \sigma_j^2 - 2\Sigma_{ij}$. Equivalently, if the report variance is written as v_i, the scored variance is $v_i + \sigma_j^2 - 2\Sigma_{ij}$ and at truthful reporting $v_i = \sigma_i^2$. This variance is positive unless the two sensors coincide almost surely.

If the mechanism instead uses a conservative or misspecified positive variance inflation, Lemma 4 implies that the elicited mean remains the conditional mean of $Y_j \mid Y_i$. Therefore, in the flat-prior honest-source model, the mean input to the WLS estimator remains Y_i even when the variance inflation is misspecified. Only the variance report is distorted. Section 5 confirms this numerically for the equicorrelated family, where the optimal mean stays within the search-grid resolution across correlations up to $\rho = 0.9$.

For general Σ, the diversity condition and the impossibility argument are unchanged because they depend only on the mean design $A = [\mathbf{1} \mid \alpha]$. The efficient estimator is the generalised least squares estimator

$$(A^\top \Sigma^{-1} A)^{-1} A^\top \Sigma^{-1} m, \tag{14}$$

while the diagonal WLS estimator (7) remains unbiased for (θ, s) under any true covariance because its rows $c_\theta^\top, c_s^\top$ satisfy

$$c_\theta^\top A = (1, 0), \qquad c_s^\top A = (0, 1). \tag{15}$$

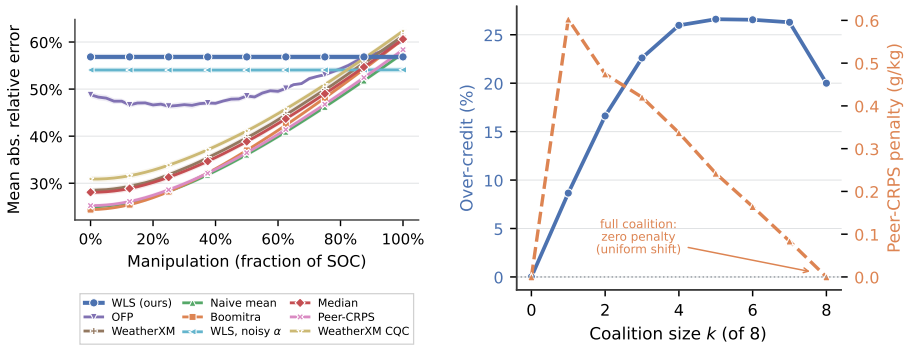
The covariance affects precision through

$$\mathrm{Var}(\hat{s}) = c_s^\top \Sigma c_s, \tag{16}$$

and similarly for $\hat{\theta}$. Common-mode covariance can reduce $\mathrm{Var}(\hat{s})$ because $c_s^\top \mathbf{1} = 0$, whereas covariance aligned with the contrast direction c_s can increase it. Thus covariance affects detection power and stake sizing, but not the algebraic identifiability condition or the unbiasedness identities.

D Coalition Manipulation

A coalition C of reporters that adds a bias vector $\boldsymbol{b}$ (supported on C) to its reported means shifts the estimate by $(\Delta\hat{\theta}, \Delta\hat{s}) = (c_\theta^\top \boldsymbol{b},\ c_s^\top \boldsymbol{b})$, linear in $\boldsymbol{b}$, where c_θ, c_s are the rows of $(A^\top W A)^{-1} A^\top W$. Because $c_\theta^\top \mathbf{1} = 1$ and $c_s^\top \mathbf{1} = 0$, a coalition that controls every sensor and adds a common bias $b\mathbf{1}$ inflates the credit by the full b while leaving $\hat{s}$, and by translation equivariance every peer score, unchanged: the all-network common shift is the only manipulation that is both undetected and unpenalised, identical to the uniform-sensitivity failure of Theorem 1. Any partial coalition ($C \neq \{1, \ldots, N\}$) moves its reports away from its peers and pays a strictly positive peer-CRPS penalty, which falls with coalition size and reaches zero only at the full network (Figure 4(b)). There is thus no hard tolerance threshold: truthful reporting degrades gracefully, and a single deviant peer perturbs an honest sensor's optimal mean by only about $b/(N-1)$ (Section 4). Sybil-resistant identity, which prevents one party from controlling the whole ensemble, confines manipulation to the penalised partial regime.



(a) (b)

Fig. 4. Baselines over-credit; collusion is penalised by the peer score. (a) Absolute error versus manipulation for all methods; the curves converge near the noise floor. **(b)** A coalition of k reporters adds a common bias to inflate the credit; every partial coalition pays a peer-CRPS penalty, and only the all-eight common shift pays zero, the collusion form of the necessity boundary.

Figure 4(a) shows the absolute errors of all methods converging near the noise floor at full manipulation.

E Regression Baselines

A regression that models the source shift as a nuisance parameter, fitting (θ, s) on the design $[\mathbf{1} \mid \boldsymbol{\alpha}]$, inherits the manipulation invariance whatever loss it

minimises. Table 5 fits (θ, s) by weighted least squares (the estimator of (7)), a Huber loss, and an ℓ_1 loss. Each holds a credit bias between -1.3% and -1.5% that is constant across the manipulation range, while the shift-naive aggregators of Table 3 reach $+57\%$. The invariance is therefore a property of the estimator structure, not of the loss or of the payments. The incentive layer does a different job: it does not lower the bias, which any shift-modelling regression already removes, but makes truthful reporting and $s = 0$ a strict equilibrium (Theorem 1) and supplies the detected $\hat{s}$ that the on-chain stake slashes (Appendix B).

Table 5. Shift-modelling regressions are manipulation-invariant. Credit bias $\mathbb{E}[(\hat{\theta} - \theta)/\theta]$ for estimators that jointly fit (θ, s), constant across 0 to 100% manipulation, against the $+57\%$ of the shift-naive baselines (Table 3).

Estimator (fits θ and s)	Credit bias
Weighted least squares (7)	-1.3%
Huber	-1.5%
ℓ_1	-1.5%

F Calibration and Deployment

Sensitivity drift. The most common calibration change, a uniform rescaling of all α_i from sensor ageing, leaves $\hat{\theta}$ exactly unbiased, because it does not change the column space of the design (Section 4). Only the non-proportional part of a drift biases the credit, and that bias is bounded by $|s|\,\|\tilde{c}_\theta\|\,\|\boldsymbol{\Delta}\|$ and stays below the baseline manipulation bias even at a 50% drift (Section 5). The sensitivities should be re-measured periodically on trusted reference soils rather than source-controlled sites, so re-calibration opens no manipulation channel.

Verifying diversity on-chain. The contract holds the published (σ_i, α_i) and accepts an ensemble only when $\operatorname{rank}(A) = 2$, equivalently when the precision-weighted variance $\sum_i w_i(\alpha_i - \bar{\alpha}_w)^2$ of (10) exceeds a floor. Sensor heterogeneity is then a verifiable procurement requirement rather than an assumption, and the same quantity sets the detection power and the stake.

Limits. The measured α_i are a single-survey snapshot. Their stability across seasons and soil textures, and a validation against paired wet-and-dry field measurements, need longitudinal data that we leave to deployment.

G Extensions

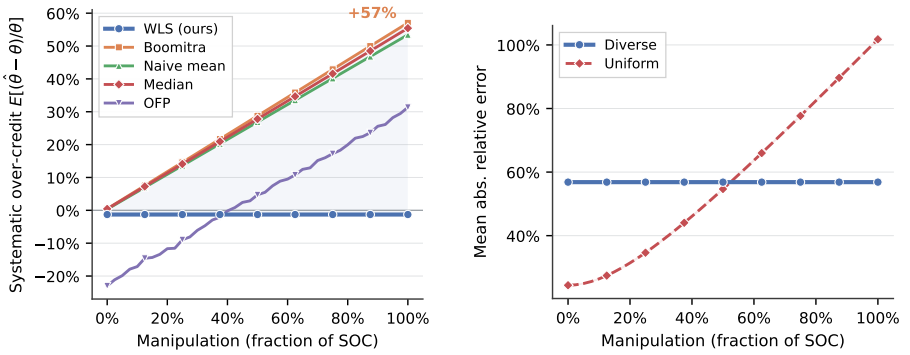
Multiple manipulation modes. With k modes $s \in \mathbb{R}^k_{\geq 0}$ and $Y_i = \theta + \sum_{j=1}^{k} \alpha_i^{(j)} s_j + \varepsilon_i$, the means stack into the design $A = [\,\mathbf{1} \mid \boldsymbol{\alpha}^{(1)} \mid \cdots \mid \boldsymbol{\alpha}^{(k)}\,] \in$

$\mathbb{R}^{N\times(k+1)}$. The same argument shows strict truthfulness is achievable if and only if $\operatorname{rank}(A) = k+1$: the all-ones vector lies outside the span of the manipulation directions and those directions are linearly independent. This requires $N \geq k+1$ sufficiently diverse sensors, and Theorem 1 is the case $k = 1$. As in the scalar case the action is one-sided, and a penalty non-decreasing in each $\hat{s}_j$ makes $s = 0$ the strict best response by the componentwise form of Lemma 3, in line with the one-sided slashing event. If a sensitivity varies with the true value or the manipulation level, the design A becomes state dependent and the characterisation applies to its local linearisation. The real-model results of Section 5 show the invariance tolerates the mild spectral nonlinearity of wetting.

Variance elicitation. The peer score (5) inflates the reported variance v_i by the common-knowledge peer noise σ_j^2, so strict propriety sets $v_i + \sigma_j^2 = \sigma_i^2 + \sigma_j^2$ and the mechanism elicits calibrated uncertainty along with the mean. The estimator (7) instead weights by the pre-calibrated σ_i^2 of Table 2, not re-elicited on-chain.

H Sensor Benchmark and Parametric Results

Figure 5 shows the manipulation resistance on the parametric $\theta + \alpha_i s$ model summarised in Section 5: the WLS bias stays near zero across the manipulation range while the baseline aggregators accumulate over-crediting, and the absolute error is flat under diverse sensitivities but grows under uniform ones.



(a) (b)

Fig. 5. Manipulation resistance. (a) Systematic bias versus manipulation fraction; baselines accumulate over-crediting while WLS stays near zero. **(b)** Absolute error versus manipulation fraction; flat under diverse sensitivities, growing under uniform ones.